



# DocsGuard

**AI document authenticity & fraud detection — for government schemes, first.**

Upload a document. Deterministic mathematics checks what mathematics can check. Claude reasons over what's left. An adversarial court stress-tests the verdict. And the system says plainly, every time, what it could not verify.

**60/60**

deterministic identifier tests pass

**5/5**

self-test checks, zero API key

**12**

forensic dossier modules per scan

**₹3.48L cr**

DBT leakage removed nationally\*

\*Identity dedup, not document forensics — see slide 3

# Aadhaar proves who you are. It does not prove what your paper says.

Most document fraud reaching Indian welfare, scholarship, and tender flows is not a stolen identity. It is a real, correctly-authenticated person submitting a document that says something untrue about them — or a document nobody real ever issued.

## Income certificate

A figure retyped over a genuine scan drops declared income under the EWS ₹8L/year threshold, or a scheme's own limit — the certificate itself can be a clean re-scan with the edit invisible at a glance.

## Caste / domicile certificate

Recreated from scratch on a word processor, or a real certificate with the category or issuing tehsil altered — MP's own layouts are simple and publicly visible to copy.

## Tender bank guarantee

A forger impersonates the issuing bank via a lookalike email domain to "confirm" a fabricated guarantee when the tender authority calls to verify it — the single highest-value forgery pattern in Indian procurement.

**Checked how, today?** One clerk, hundreds of applications, seconds per file, no forensic tooling. A competent forgery passes. An honest applicant whose scan just looks unusual gets rejected. Both failures are invisible until an audit finds them years later. Source: docs/FRAUD\_TYPOLOGIES.md §1, §6

# Aadhaar/DBT solved a different problem — and solved it at national scale

The Direct Benefit Transfer programme is India's flagship anti-leakage system. It works by **identity deduplication** — proving a beneficiary is a real, unique person and removing ghosts and duplicates. It was never designed to check whether an uploaded certificate's claims are true.

**5.2 cr**

fake ration cards purged nationally

IMPACT.md §1 / DBT Bharat

**4.1 cr**

duplicate LPG connections removed

IMPACT.md §1 / DBT Bharat

**2.1+ cr**

ineligible beneficiaries de-listed

IMPACT.md §1 / DBT Bharat

**1.5bn+**

non-unique DBT transactions / year, 320+ schemes

IMPACT.md §1

## What identity dedup cannot catch

- **~26% of scholarship applications flagged fake** in the 2022–23 national cycle — 6.7+ lakh of 25.5 lakh, a document/eligibility fraud rate, not an identity-duplication one.
- **1,000+ MP government employees** are under investigation for fraudulent caste certificates, including **600+ Class I officers** — most cases still open, most accused still in post.
- Every real person in these cases already passed Aadhaar identity checks. The document, not the person, was the fraud.

**This is exactly DocsGuard's target gap** — the document is fabricated or altered even though the applicant is a real, unique, Aadhaar-verified person.

# This has already happened, repeatedly, at multi-crore scale

| CASE                                      | AMOUNT     | PATTERN                                                                                                                                                    | SOURCE                                  |
|-------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| DBT cumulative leakage removed            | ₹3.48 L cr | National, Aadhaar-linked identity dedup — <i>not</i> document forensics; cited to show scale, not claimed as DocsGuard's addressable number.               | IMPACT.md §1                            |
| Ahmedabad fake-ITC (GST) scam             | ₹1,500 cr  | Single Dec-2025 bust; part of ₹36,374 cr detected nationally in FY24-25 alone, ₹1.14L cr cumulative 2020-25.                                               | FRAUD TYPOLOGIES.md §6.2                |
| MP Jal Nigam Maryadit fake bank guarantee | ₹202 cr    | ED case — forged PNB/Bank of Baroda guarantees submitted to MPJNL and Rajasthan Renewable Energy Corp to win contracts.                                    | PITCH_DECK.md / The420.in               |
| Fake-guarantee racket, Kolkata syndicate  | ₹183 cr    | CBI arrests incl. a serving PNB official; the same reported case is tied by FRAUD TYPOLOGIES.md to an MPJNL tender that misled ₹974 cr in contract awards. | PITCH_DECK.md / Angel One               |
| Himachal Pradesh scholarship scam         | ₹181 cr    | CBI case, 2013-2017, fake SC/ST/OBC claims; one institute alone listed 800-1,000 non-existent students to draw ₹20-25 cr.                                  | FRAUD TYPOLOGIES.md §6.3                |
| Minority scholarship scheme misuse        | ₹144.83 cr | 830 of 1,572 reviewed minority institutions found fake or inactive, over five years.                                                                       | IMPACT.md §1 / FRAUD TYPOLOGIES.md §6.3 |

**We deliberately do not overclaim this.** No consolidated "₹X crore lost to fake documents in [state] welfare schemes per year" figure exists publicly, and we will not invent one. The table above is six independently reported, sourced cases — not a single audited total, and not DocsGuard's own measured result. DocsGuard has not yet run against real government volumes.

# Upload a document. Get a defensible verdict.

- **Verdict** — Authentic / Suspicious / Likely fake, with a 0–100 risk score and confidence
- **Red flags** — each quoting the exact evidence on the document, ranked by severity
- **Consistency checks** — PASS / FAIL / WARN cross-field and arithmetic logic tests
- **Identifier validation** — Aadhaar, PAN, GSTIN, IFSC, IBAN, MRZ checksums, computed in code
- **12-module forensic dossier** — executive, forensics, typography, content, identity, security, issuer, financial, compliance, screening, predictive, verdict
- **externalChecksNeeded** — what it explicitly could NOT verify, never hidden, never invented

## SAMPLE VERDICT CARD

**LIKELY FAKE** risk 88 · confidence 74

**Income figure re-typed** — font mismatch + compression seam over the wiped region

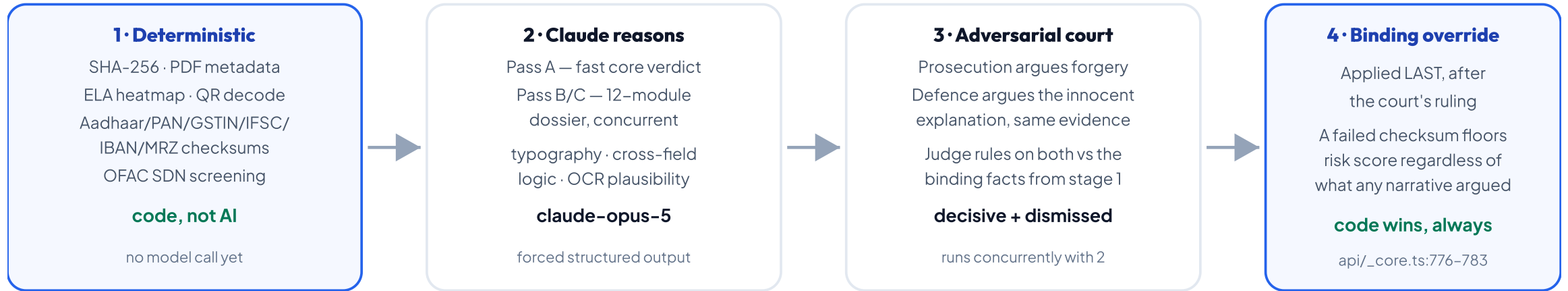
**PDF modified 41 min after creation** — re-opened and re-saved after issue

**Declared income contradicts bank statement** in the same submission bundle

**Identifier fails checksum** — mathematically impossible to have been issued

externalChecksNeeded: confirm certificate number against MP e-District portal

# Four stages. Code goes first and code goes last.



**The order is the product.** Anything a computer can prove, a computer proves — first and last — and neither Claude's reasoning nor the court's ruling is permitted to argue with it. Source: [api/\\_core.ts](#), [api/court-fast.ts](#)

# One tampered income certificate, stage by stage

A clerk uploads a scanned income certificate. The applicant photo-edited the income figure to fall under a scheme's eligibility threshold before rescanning. Here is exactly which signal fires at which stage, and why the file model traces this back to real, documented forgery patterns for this document class. *Signal types: FRAUD\_TYPOLOGIES.md §1*

|                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>STAGE 1 · DETERMINISTIC</div> <div>Two flags fire in code</div> <ul style="list-style-type: none"><li>PDF modified 41 min after creation — &gt;60s gap flag, re-opened and re-saved post-issue</li><li>Producer string = a consumer photo-editing app, not the tehsil e-Gov CMS</li><li>ID numbers on this document pass their format checksums — no hard override triggers here</li></ul> | <div>STAGE 2 · CLAUDE VISION</div> <div>Pixel-level tell</div> <ul style="list-style-type: none"><li>Font mismatch on the income figure vs. the template's fixed text</li><li>Compression seam — the edited region shows different JPEG blockiness than the surrounding scan</li><li>Consistency check FAIL — declared income contradicts the bank statement in the same bundle</li></ul> | <div>STAGE 3 · THE COURT</div> <div>Argued both ways</div> <ul style="list-style-type: none"><li>Defence: "could be a routine rescan" — Weak, no corroborating benign explanation for the font mismatch</li><li>Prosecution: font seam + compression seam + income/bank contradiction, three independent signals agreeing — Strong</li><li>Judge: decisive = the three-signal agreement; dismissed = "routine rescan" as insufficient against it</li></ul> | <div>STAGE 4 · RESULT</div> <div>Verdict issued</div> <ul style="list-style-type: none"><li><div>LIKELY FAKE</div></li><li>risk score 82, confidence 77</li><li>externalChecksNeeded: confirm the certificate number against the e-District portal directly — never via a contact on the document itself</li></ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# What mathematics decides before any model sees the file

| IDENTIFIER       | RULE APPLIED                      | A FAIL PROVES                                |
|------------------|-----------------------------------|----------------------------------------------|
| Aadhaar          | Verhoeff checksum algorithm       | Number could not have been issued            |
| PAN              | Structure + entity-type code      | Malformed / fabricated PAN                   |
| GSTIN            | mod-36 check character            | Invalid registration number                  |
| IFSC             | Structural format validation      | Not a real bank/branch code shape            |
| IBAN             | mod-97 checksum                   | Account number fails international standard  |
| MRZ ( ICA0 9303) | TD1 / TD2 / TD3 check-digit rules | Passport/ID machine-readable zone is invalid |

Source: api/verification.ts; module narrative in api/\_core.ts:736. Algorithms confirmed against the live code, not recalled from memory.

60/60

unit tests pass, verified live

```
node --test api/verification.test.mjs
```

→ 60 pass, 0 fail

**Why this matters:** a checksum FAIL is mathematical proof — not a judgment call, not something a persuasive AI argument or an adversarial court ruling can override. It is the one signal in the whole system that is 100% deterministic, reproducible, and binding.

# We made the AI argue with itself

A single AI pass either cries fraud at every compression artifact, or waves everything through. So every scan runs a three-role adversarial review, in one structured call, concurrently with the 12-module dossier.

## Prosecution

Builds the strongest honest case that the document is forged — citing only evidence actually visible in the image, the ELA heatmap, or the verified metadata. Weak points are dropped, not padded.

## Defence

Argues the innocent explanation first — JPEG recompression, scanner noise, a legitimate re-save — and flags exactly where the prosecution overreaches.

## Judge

Weights both against the **binding** code-computed facts from Stage 1. Records what was **decisive** and what was **dismissed** — and why.

**Then the binding hard-failure override runs one more time, after the ruling.** No prosecution case, no defence argument, no judge's narrative can undo a failed checksum or a sanctions list that failed to download — the floor is re-asserted last, in code, unconditionally. Source: [api/\\_core.ts:776-783](#), [api/court-fast.ts](#)

# What we actually ran, today, on this code

| CLAIM                                         | EVIDENCE                                                                                                                                                 |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier validation is real mathematics     | 60/60 unit tests pass — Verhoeff, PAN, GSTIN mod-36, IFSC, IBAN mod-97, ICAO 9303 MRZ (TD1/TD2/TD3) — <code>node --test api/verification.test.mjs</code> |
| The deterministic engine runs with no API key | <code>npm run selftest</code> → 5/5 checks pass: synthetic doc generation, ELA routine, ELA sanity, QR round-trip, verification-suite run                |
| ELA responds to a real pixel edit, not noise  | mean error 0.196 → 0.264 when a splice was introduced into the same test image — measured, deterministic, reproducible                                   |
| Watchlist screening is live, not mocked       | Real US Treasury OFAC SDN CSV export fetched live (HTTP 200) and parsed — code enforces a >1,000-record sanity floor before trusting the result          |

**And what we refuse to claim:** Error-Level Analysis is weak on flat, vector-rendered documents — ordinary body text produces as much error as a splice (see next slide). We report it as one indicative signal among several, never as a standalone verdict — and we would rather tell you that than be caught by a reviewer who knows it.

## What DocsGuard cannot yet do — stated plainly

**ELA is weak on flat, vector-rendered documents.** ELA depends on a prior JPEG compression history to diff against. A PDF rendered to PNG, a vector-drawn certificate, or an already high-quality JPEG has little compression history for a single resave to reveal — the whole image reads as uniformly low-error and a spliced region may not stand out.

**No live source-of-truth is wired up yet.** No DigiLocker, e-District, UIDAI, GST/PAN registry, or MCA call happens today. Anything that needs one is listed in `externalChecksNeeded`, never asserted — but a well-forged, internally-consistent document can still pass as AUTHENTIC.

**Confidence is self-reported, not statistically calibrated.** It is the model's own stated certainty — not checked against a labeled dataset, a held-out test set, or repeated sampling. Treat it as a triage signal, not a validated probability.

**PDF metadata is itself editable.** Producer, creator, author, and timestamp fields can be spoofed by a forger using a metadata-editing tool. They are genuinely useful — most casual forgers don't bother cleaning them — but they are evidence, not proof.

Source: docs/FORENSIC\_METHODODOLOGY.md §6, items 1, 3, 4, 6.

# The obvious open-source shortcuts are not actually free

We researched the three most-cited open-source ID/KYC/signature-verification repos before building anything in-house. All three ship **no LICENSE file** — under default copyright law, that means no permission to use, modify, or redistribute, not "MIT-like."

| PROJECT                                 | WHAT IT DOES                                     | LICENCE — AS REPORTED BY GITHUB'S OWN API                                                                                                                                                              |
|-----------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MiniAiLive/ID-DocumentRecognition       | ID/passport/driver's-license OCR, 200+ countries | <b>license: null</b> — commercial, trial-gated; "Contact us to get a trial License." Windows-only.                                                                                                     |
| DoubangoTelecom/KYC-Documents-Verif-SDK | ID/passport/visa recognition, 250+ countries     | <b>license: null</b> — model weights in a private, approval-gated repo; decompiling forbidden by the README.                                                                                           |
| amaljoseph/...YOLOv5-and-CycleGAN       | Signature detection, cleaning, verification      | <b>license: null</b> — no grant at all, <b>and</b> its YOLOv5 dependency is Ultralytics-licensed <b>AGPL-3.0</b> , forcing full source disclosure or a paid Enterprise licence for any commercial use. |

**What this means for DocsGuard:** the deterministic checksum layer (Verhoeff/PAN/GSTIN/IFSC/IBAN/MRZ) and ELA are hand-built in TypeScript with zero new dependencies and zero licensing exposure — the closest genuinely reusable reference we found was MIT-licensed but Python/PyTorch (camilooscargbaptista/cv-fraud-detection), useful to validate our architecture direction, not to vendor. Source: docs/COMPETITIVE\_RESEARCH.md §1, §3

# What we can connect to — and what it really costs

| SOURCE                             | WHAT IT PROVES                           | REALITY                      |
|------------------------------------|------------------------------------------|------------------------------|
| Aadhaar offline QR / XML           | Signed identity data vs UIDAI public key | Buildable now                |
| OFAC / UN / EU sanctions lists     | Watchlist screening                      | Live today                   |
| ICAO 9303 MRZ                      | Passport / ID check digits               | Shipped                      |
| PAN / GSTIN / MCA / IEC registries | Registration existence and status        | Paid aggregator required     |
| DigiLocker / e-District            | Did this office actually issue it?       | Needs government partnership |

DigiLocker is the endgame — it turns *"internally consistent"* into *"issued by that office."* MP already has live e-District DigiLocker integration for some document classes (e.g. RCMS land records); others, like older marksheets, reportedly fail to resolve reliably. It needs a government sponsor — exactly what a department pilot unlocks. Source: docs/IMPACT.md §4 (Phase 2)

# One scheme, one office, measured against manual review

### Phase 1 — pilot

One high-fraud-signal scheme, one district office, 3–6 months. Runs alongside existing manual review as decision support — every verdict advisory, no auto-rejection. This is the only way to measure real precision/recall against outcomes officers already have to determine anyway.

### Phase 2 — district-wide

Full document set at one office: caste/income/domicile certificates, marksheets, PDS bank statements. Begin turning externalChecksNeeded items into real PASS/FAIL via e-District/DigiLocker where a certificate carries a resolvable application number.

### Phase 3 — state-wide

Roll out to all districts for the schemes validated in Phase 1–2; add the officer dashboard (queue, filter by verdict/severity, audit trail); evaluate extension to tender-document screening.

## What the pilot must measure before any scale decision

### Fraud-catch rate

of flagged docs, % confirmed fraudulent on investigation

### False-positive rate

the single most important trust metric — tracked from day one

### Officer time/doc

decision time, not raw API latency

### Turnaround time

submission to decision, before vs after

### Coverage

% of scheme volume actually routed through DocsGuard

Source: docs/IMPACT.md §3, §4. None of these numbers exist for DocsGuard yet — they are the deliverable of Phase 1, not a pre-launch claim.

# One scheme. One department. Real documents.

A pilot gives us the two things we cannot generate ourselves: a labelled dataset to calibrate confidence against, and the government sponsorship needed for DigiLocker issuer verification.

## Non-negotiables

- Counter-level triage — clerk uploads, gets a reason in seconds
- Every verdict stores its full evidence — a real audit trail
- **The officer decides.** DocsGuard recommends; it never auto-rejects

## What we bring on day one

- A working, tested pipeline — 60/60 identifier tests, 5/5 selftest, live OFAC screening
- An adversarial court that shows its work — decisive and dismissed, every time
- An honesty architecture that never fabricates a lookup it didn't perform

[github.com/Yashraj00700/pramaan](https://github.com/Yashraj00700/pramaan)



# Proof, in seconds, that doesn't lie about what it knows.

Deterministic mathematics decides what mathematics can decide. Claude reasons over what's left. An adversarial court stress-tests the outcome. And the officer decides — always.

[github.com/Yashraj00700/pramaan](https://github.com/Yashraj00700/pramaan) · Thank you — questions welcome.